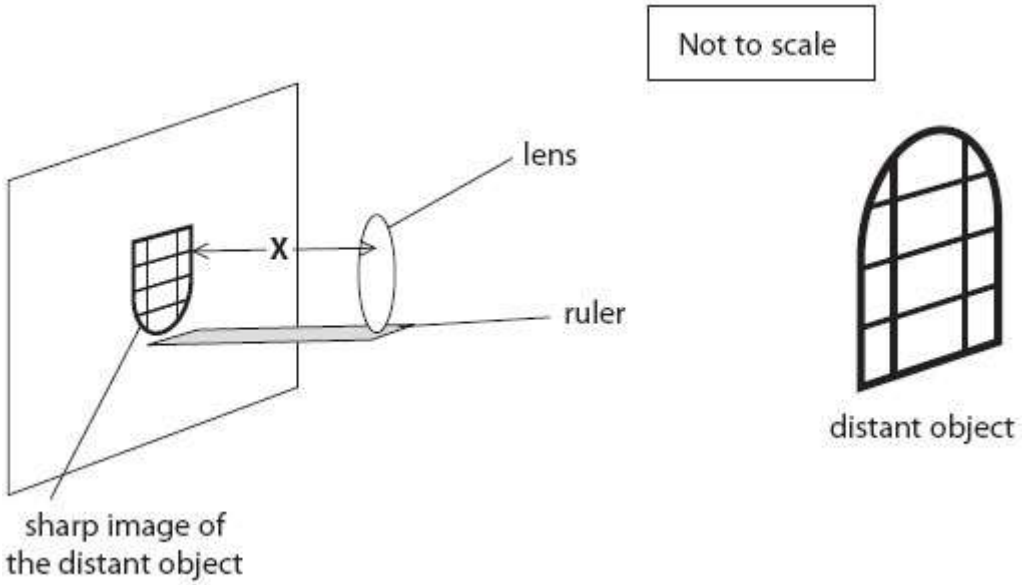
Questions

Q1.

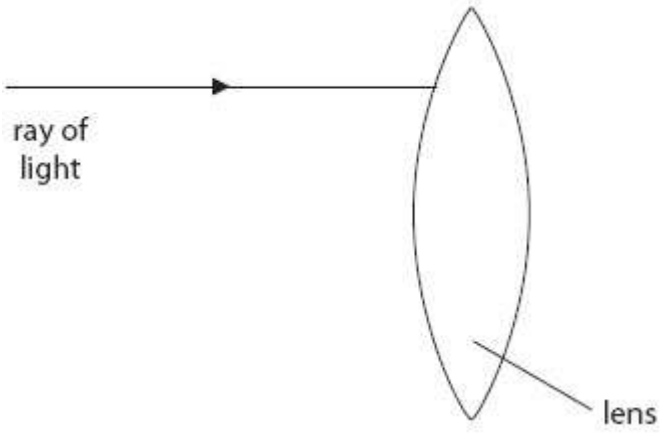
(a) Some students investigate a converging lens.
The students set up the apparatus as shown.



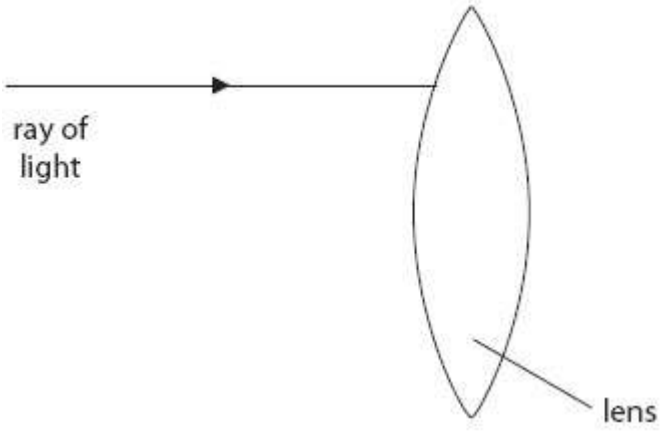
Complete the sentence by putting a cross (■) in the box next to your answer.

(i) The distance **X** is

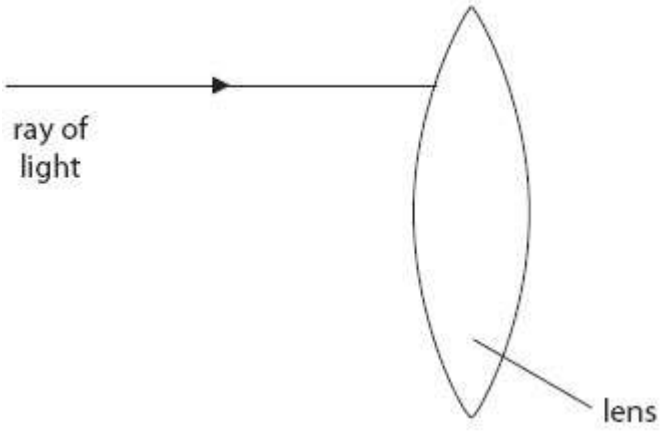
(1)



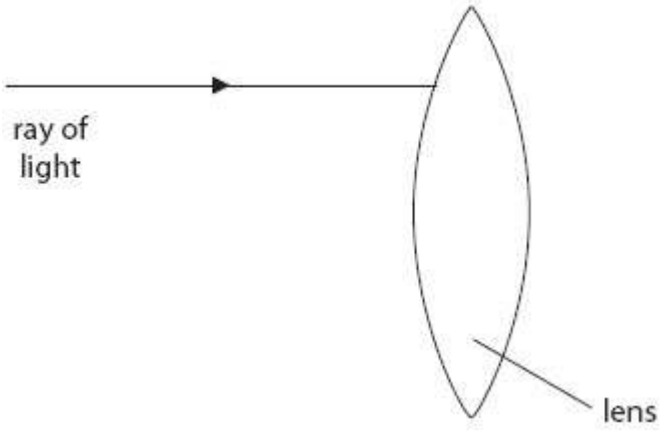
A the focal length



B the object distance



C the eyepiece distance



D the magnification

(ii) Use words from the box to complete the sentences.

(2)

bigger than	diverging	real
smaller than	the same as	virtual

The size of the image is .....the distant object.
The type of image formed on the screen is a .....image.

(b) The diagram shows a ray of light as it arrives at a lens.

Draw the path of the ray inside the lens.

(1)

(c) The students use a telescope to view the Moon.

Light from the Moon takes 1.3 s to reach the students.

The speed of light is 300 000 km/s.

Calculate the distance to the Moon.

(2)

distance = speed × time

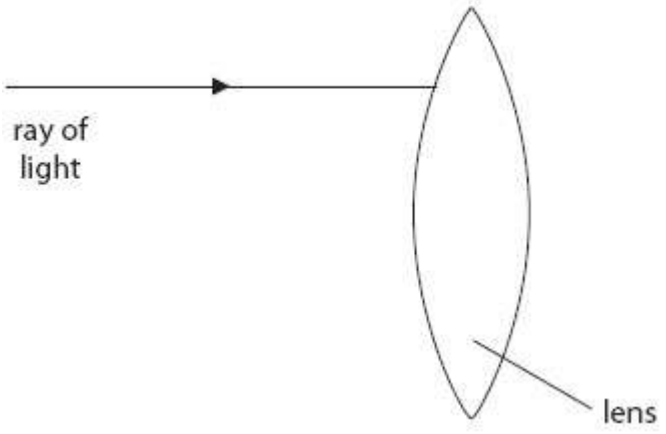
.....

(d) Complete the sentence by putting a cross (■) in the box next to your answer.

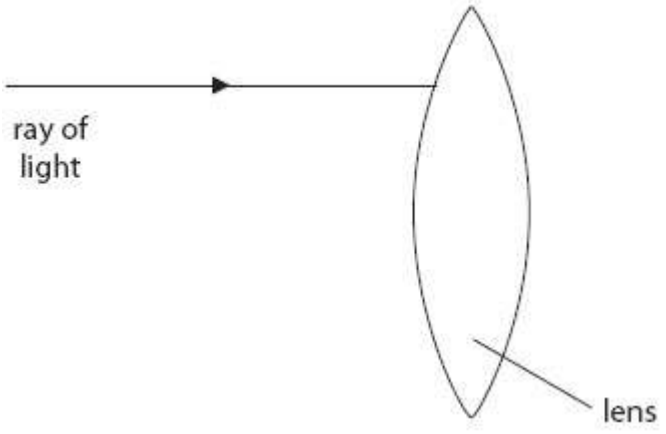
A satellite orbits the Moon.

Radio waves from this satellite transfer

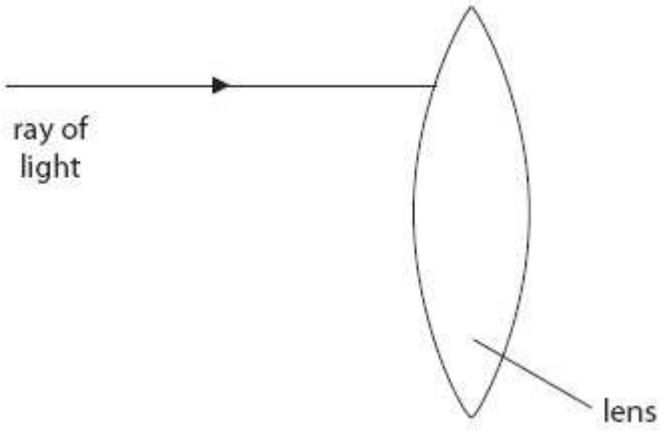
(1)



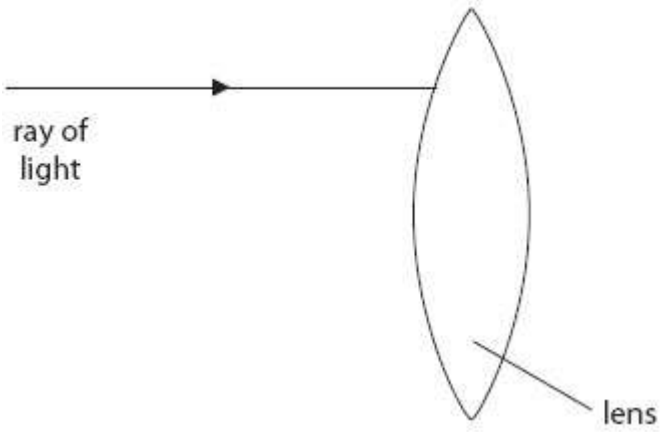
A matter only



B energy and matter



C information and matter



D energy and information

(Total for Question is 7 marks)

Q2.

Figure 1 shows a speed/time graph for a car.

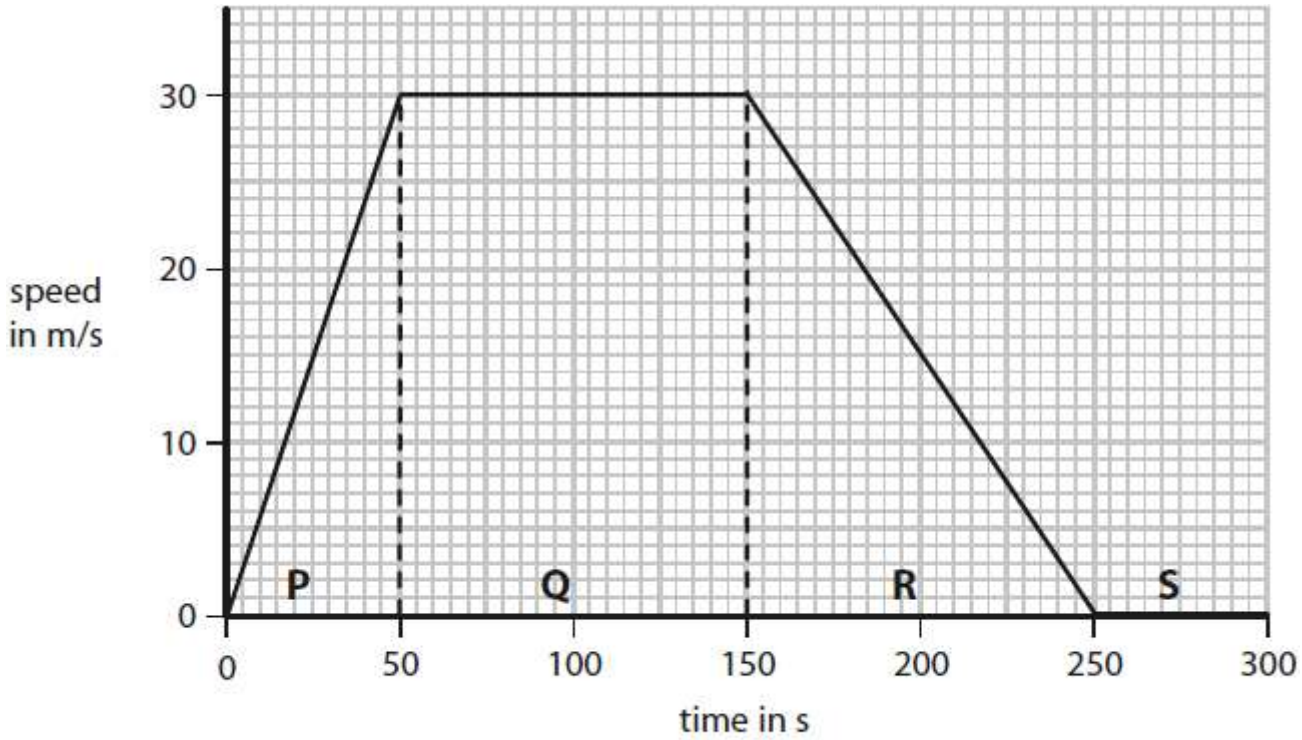


Figure 1

(i) The graph in Figure 1 is divided into four parts, **P**, **Q**, **R** and **S**.

Draw a line from the letter for each **part** to the correct **description of the motion** during that part.
One line has been drawn for you.

(2)

part	description of the motion
P	the car is standing still
Q	the car is accelerating
R	the car is decelerating
S	the car is travelling at constant speed

(ii) In two parts of the graph in Figure 1 the forces are balanced.

State the letters of the two parts of the graph where the horizontal forces acting on the car are balanced.

(2)

part ..... and part .....

(iii) Calculate the distance travelled by the car in part Q.

Use the equation

distance travelled = average speed × time

(2)

distance travelled = ... m

(Total for question = 6 marks)

Q3.

Shot-put is an Olympic event.
The shot is a heavy ball.
An athlete throws the shot as far as possible.

A sports scientist analyses an athlete's throw to help improve performance.

The scientist can measure several quantities in the analysis.

Which one of the following is a scalar quantity?

(1)

☐ **A** acceleration

☐ **B** force

☐ **C** mass

☐ **D** velocity

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these is a vector?

(1)

☐ **A** energy
☐ **B** force
☐ **C** mass
☐ **D** work

(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these is a scalar quantity?

(1)

☐ **A** acceleration
☐ **B** distance
☐ **C** force
☐ **D** weight

(Total for question = 1 mark)

Q6.

(i) Which of these would be a typical speed for a racing cyclist travelling down a steep straight slope?

(1)

☐ **A** 0.2 m/s

☐ **B** 2 m/s

☐ **C** 20 m/s

☐ **D** 200 m/s

(ii) A cyclist travels down a slope.
The top of the slope is 20 m vertically above the bottom of the slope.
The cyclist has a mass of 75 kg.

Calculate the change in gravitational potential energy of the cyclist between the top and the bottom of the slope.

The gravitational field strength, g, is 10 N/kg.

(3)

change in gravitational potential energy = ... J

(Total for question = 4 marks)

Q7.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these is a typical human reaction time?

(1)

☐ **A** 0.0027 s
☐ **B** 0.027 s
☐ **C** 0.27 s
☐ **D** 2.7 s

(Total for question = 1 mark)

Q8.

(i) Which row of the table shows the correct units of weight and of mass?

(1)

		unit of weight	unit of mass
☐	**A**	kilogram	newton
☐	**B**	newton	kilogram
☐	**C**	newton per kilogram	newton
☐	**D**	newton per kilogram	kilogram

(ii) State the name of a piece of equipment that can be used to measure mass.

(1)

.....

(Total for question = 2 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(i) Which of these is the correct equation that relates force, mass and acceleration?

(1)

☐ **A** $F = m + a$
☐ **B** $F = m - a$
☐ **C** $F = m \times a$
☐ **D** $F = m \div a$

(ii) A cyclist has a mass of 70 kg.

Calculate the force needed to accelerate the cyclist at 2.0 m / s^2.
State the unit.

(2)

force = ... unit = ...

(Total for question = 3 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graph in Figure 5 shows how the velocity of a car changes with time.

The car starts from rest and travels along a level, straight road for 50 s.

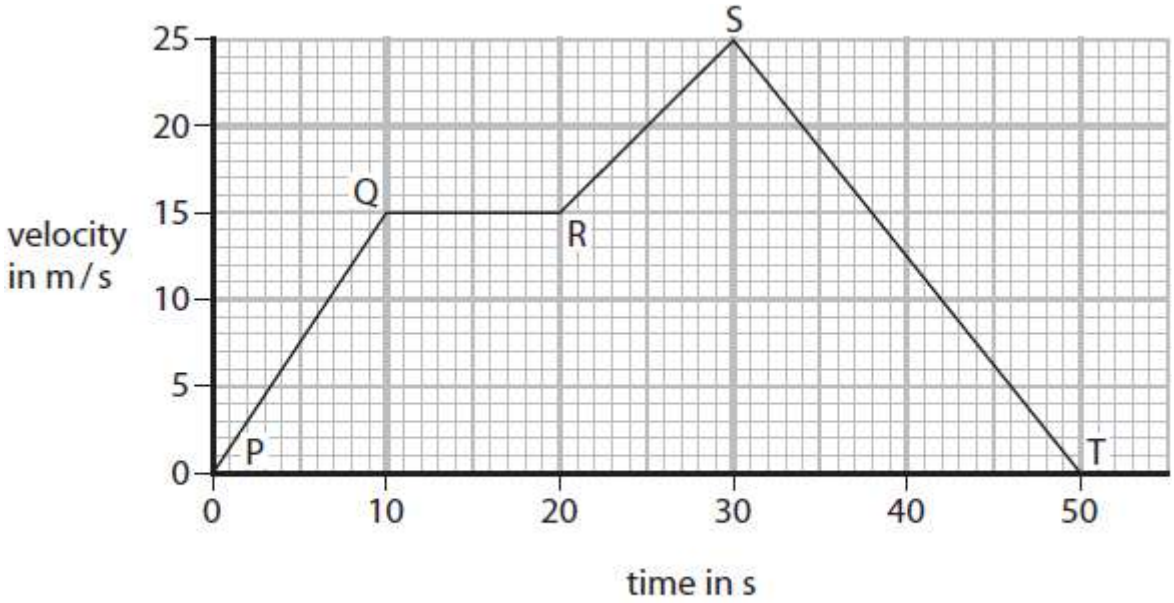


Figure 5

(i) Which part of the graph shows when the car has constant velocity?

(1)

☐ **A** PQ
☐ **B** QR
☐ **C** RS
☐ **D** ST

(ii) Which part of the graph shows when the car has the greatest acceleration?

(1)

☐ **A** PQ
☐ **B** QR
☐ **C** RS
☐ **D** ST

(iii) Calculate the acceleration of the car in the first 10 s shown on the graph.

(2)

Use the equation

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

acceleration = ... m/s^2

(iv) Calculate the distance the car travels in part QR shown on the velocity / time graph in Figure 5.

(3)

distance = ... m

(Total for question = 7 marks)

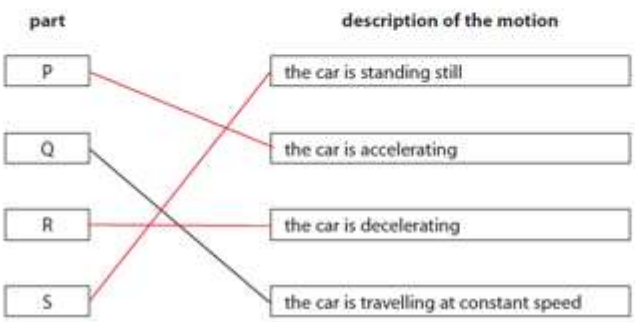
Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
(a)(i)	A the focal length (1)		**(1)**
(a)(ii)	smaller than (1) real (1)		**(2)**
(b)	• Any (more or less) straight ray which changes direction inside the lens (1)	Ray does not need to touch far side. Allow slight discontinuities Ignore any ray drawn beyond the 2nd surface and any reflected ray(s). Ignore any extra incident rays.	**(1)**
(c)	substitution into given equation (1) 1.3 × 300 000 evaluation (1) 390 000 (km)	Power of 10 error max 1 mark 3.9×10^5 (km) 2 marks for correct numerical answer with no working shown Ignore any unit given by candidate.	**(2)**
(d)	D energy and information (1)		**(1)**

Total for Question = 7 marks

Q2.

Question Number	Answer	Mark
(i)	all three correct (2) one or two correct (1)  	**(2)**

Question Number	Answer	Additional guidance	Mark
(ii)	Q and S Q (1) (and) S (1) OR S (1) (and) Q (1)	in either order maximum of 1 mark if 3 letters given no marks if 4 or more letters given	(2)

Question Number	Answer	Additional guidance	Mark
(iii)	substitution (1) (distance =) 30 x 100 evaluation (1) 3000 (m)	for 1st mp accept 100 x 30 OR (30 x 50) x 2 award full marks for the correct answer without working allow **1 mark** for EITHER 30 x 50 OR 30 x 150 OR 30 x 250	(2)

Q3.

Question number	Answer	Mark
	C mass	(1)

Q4.

Question number	Answer	Mark
	☒ **B** force Options A, C and D are all scalars.	**(1)**

Q5.

Question Number	Answer	Additional guidance	Mark
	B distance **A**,**C**, and **D** are incorrect as these are vector quantities		**(1)** **AO1**

Q6.

Question Number	Answer	Mark
(i)	The only correct answer is **C 20 m/s** A is not correct because 0.2 m/s is too slow B is not correct because 2 m/s is too slow D is not correct because 200 m/s is too fast	**(1)**

Question Number	Answer	Additional guidance	Mark
(ii)		NO PoT error NO ecf from wrong equation	(3)
	recall (1) (ΔGPE) = m × g × Δh	mgh or m x g x h	
	substitution (1) (ΔGPE =) 75 x 10 x 20	75 x 10 x 20 scores the first 2 marks	
	evaluation (1) 15 000 (J)	accept 14700 (J) from using g = 9.8 (N/kg) award full marks for the correct answer without working	

Q7.

Q8.

Q9.

Question number	Answer	Mark
(i)	☒ **C** $F = m \times a$ A, B and D have incorrect mathematical operator	(1)

Question number	Answer	Additional guidance	Mark
(ii)	140 (1) N (1)	no ecf from 2ai independent mark allow newton(s) n do **not** allow Ns ns	(2)

Q10.

Question Number	Answer	Mark
(i)	**B** QR (horizontal line) **A** PQ is incorrect it shows constant acceleration **C** RS is incorrect it shows constant acceleration **D** ST is incorrect it shows constant deceleration	(1) AO3

Question Number	Answer	Mark
(ii)	**A** PQ (steeper slope shows greater acceleration) **B** QR is incorrect it shows zero acceleration **C** RS is incorrect as slope is less steep than for PQ **D** ST is incorrect as the slope is less steep than for PQ and shows deceleration	(1) AO3

Question Number	Answer	Additional guidance	Mark
(iii)	substitution (1) (a=) $\frac{15(-0)}{10}$ evaluation (1) 1.5 (m/s^2)	15 seen Allow 10 divided by any number between 6 and 7 for this mark award full marks for the correct answer with no working	(2) AO3

Question Number	Answer	Additional guidance	Mark
(iv)	indication that distance travelled = area under graph (1) substitution (1) (distance travelled =) 10 x 15 evaluation (1) 150 (m)	may be seen on graph accept (distance=)speed x time ignore speed = $\frac{\text{distance}}{\text{time}}$ award full marks for the correct answer with no working award 2 marks for 10 x 15 seen anywhere if no other marks awarded, 1 mark for use of 15 (m/s) or 10 (s)	(3) AO3